

Lab 3: Component Modelling & Architectural Pattern Selection

Self-Service Coffee Kiosk System

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Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

The system separates concerns into a Presentation Layer (touchscreen ordering interface), a Business Layer (order management, payment processing, receipt printing logic), and a Data Layer (menu and pricing storage). Each layer depends only on the layer beneath it, giving a clear, maintainable structure appropriate for a single, self-contained kiosk device.

Reason 1: Matches the Scale and Deployment Model of a Single Kiosk

A coffee kiosk is a single-purpose, self-contained device handling one customer transaction at a time, not a distributed, internet-scale system. Layered architecture provides exactly the structure needed for this scope: a straightforward request flow from touchscreen input to order logic to hardware and data access, without the operational overhead of independently deployed services (as in Microservices) that this scenario does not require.

Reason 2: Simplifies Integration of Fixed Hardware and a Small, Stable Menu

The kiosk has a fixed, well-known set of responsibilities: 3 coffee types, 2 sizes, one payment method, and one receipt printer. These requirements are unlikely to change frequently. A layered design lets the Business Layer coordinate the Payment Service and Receipt Printer components through clearly defined interfaces, while the Data Layer keeps menu and pricing details isolated and easy to update without touching the ordering logic.

Security Advantage

Layering enforces that the Presentation Layer never talks directly to the Data Layer or external payment hardware; all requests must pass through the Business Layer's Order Manager and Payment Service components. This chokepoint allows credit-card transaction data to be validated, tokenized, and access-controlled in one place, reducing the attack surface compared to a design where the UI could bypass business rules and query sensitive data directly.

Performance Benefit

Because each layer has a single, well-defined responsibility, the Business Layer can cache frequently accessed menu and pricing data retrieved from the Data Layer, avoiding a repeated database query for every touchscreen interaction. This keeps the customer-facing ordering flow fast and responsive, which matters in a busy café where order throughput directly affects customer wait times.

Component Overview

- Kiosk / Touchscreen UI Component (Presentation) — handles customer coffee-type, size, and payment-trigger selection via the touchscreen; sends orders through the Order Interface.
- Order Manager Component (Business) — coordinates the order lifecycle: receives orders from the UI, requests payment authorization, triggers receipt printing, and looks up menu/pricing data.
- Payment Service Component (Business) — processes credit-card payment authorization for an order and returns an approval/decline result to the Order Manager.
- Receipt Printer Component (Business) — formats and sends order details to the physical receipt printer hardware once payment is confirmed.
- Menu & Pricing Database Component (Data) — stores the coffee menu, sizes, and prices, and answers lookup queries from the Order Manager.

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UML Component Diagram — Layered Architecture

